The Injective Generation Condition

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Introduction

The objective of the talk:

1. Introduce quiver algebras and their representations, and a pictorial way to represent its submodules and quotient modules as a module over itself.
2. Introduce triangulated (sub)categories and their relation to the derived category $\mathscr{D}(R)$.
3. Introduce the injective generation condition and basic methods to show that injective generates for an algebra.

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1. Quiver Algebras
2. Quiver Representations
3. Some Triangles...
4. Injective Generation

Quiver Algebra

Definition

A **quiver** Q is a directed graph with loops and multiple edges between each vertex allowed.

In this talk, all quivers will have finitely many vertices and edges (i.e. $\#VQ, \#EQ < \infty$.)

Quiver Algebra (2)

An algebra can be built (over an algebraically complete field k) using two data:

1. Basis: All (finite) paths in Q;
2. Multiplication of basis elements: The concatenation of paths, with noncompatible paths, is evaluated to 0.

Definition

We will denote the algebra built as kQ.

Quiver Algebra (3)

Some remarks on kQ:

- "Paths" are allowed to have only one vertex (length 0). This yields the **constant path at vertex** i, denoted by e_i.
- This is an algebra with multiplicative identity $1 := \sum_{n=1}^{\#VQ} e_n$
- kQ is infinitely dimensional if (and only if) there is a cycle in Q.

Ideals of Quiver Algebra

Definition

Given $k \in \{0, 1, 2 \cdots\}$, define the ideal I^k to be the subspace of kQ which is spanned by all paths of length at least k.

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Definition

An ideal I in kQ is **admissible** if there exists $\ell \geq 2$ such that

$$I^\ell \subseteq I \subseteq I^2$$

Why is kQ/I important?

Theorem (Gabriel, 1973)

Let A be a finite dimensional k-algebra, then there exists a quiver Q with an admissible ideal I such that

$$\text{mod-}A \simeq \text{mod-}kQ/I$$

are equivalent categories.

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Representation Theory of kQ

Definition

A **representation** of Q consist of the following two data:

- At each vertex i, a k-vector space V_i.
- At each edge $E := (v_i, v_j)$, a (k-)linear map $\phi_E : V_i \to V_j$.

Given a finite path as an edge sequence $p = (E_1, E_2 \cdots E_n)$, we shorthand

$$\phi_P := \phi_{E_n} \circ \cdots \phi_{E_2} \circ \phi_{E_1}$$

A representation (V_i, ϕ_E) of Q is **bounded by** I if for all $p \in I$ we have $\phi_p = 0$.

Given kQ, the representations of kQ bounded by I form a category $\mathrm{Repn}(kQ/I)$.

Representation Theory of kQ (2)

Theorem

$$\mathrm{Repn}(kQ/I) \simeq \text{mod-}kQ/I$$

are equivalent as categories.

Proof.

Given representation (V_i, ϕ_E) of Q bounded by I, assign it the kQ/I-module $\bigoplus_{n=1}^{\#VQ} V_n$ with right multiplication $v_n \cdot p := \phi_p(v_n)$. □

Visualizing Submodules - Motivation

Write $A := kQ/I$ for now, where Q has n vertices. Note that we have a projective decomposition (as A-modules)

$$A_A := e_1A \oplus e_2A \oplus \cdots e_nA$$

How can we see the representation more concretely in $\mathrm{Repn}(kQ/I)$?

- The class $S(i)$, at vertex j, equip the vector space k if and only if $i = j$, and equip 0 otherwise. For any oriented edge in Q, assign the zero map.
- The class $P(i)$, at vertex j, equip the vector space with the set of equivalence classes

$$\{p + I : p \text{ is a path from } i \text{ to } j\}$$

as basis. For an oriented edge α from j to k, we assign the linear map by specifying a path p will be sent to $p\alpha$ by concatenation.
- The class $I(i)$, at vertex j, equip the vector space with the set of equivalence classes

$$\{p + I : p \text{ is a path from } j \text{ to } i\}$$

as basis. For an oriented edge α from j to k, we assign the linear map by specifying a path p will be sent to the following:
  - If the path p is in the form of $\alpha p'$ (i.e. it starts with the entry α and have remaining part as a path p'), send it to p'.
  - Otherwise, send the path p to 0.

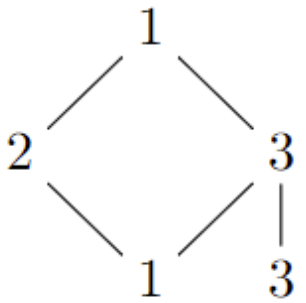
The Punchline

If we identify $S(i), P(i)$ and $I(i)$ under the equivalence introduced,

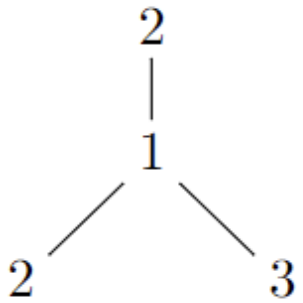
Theorem

$S(i)$ are simple, $P(i)$ are isomorphic to the projective A-module e_iA, and $I(i)$ are isomorphic to the injective A-module $\nu(Ae_1)$ (Where ν is the Nakayama functor).

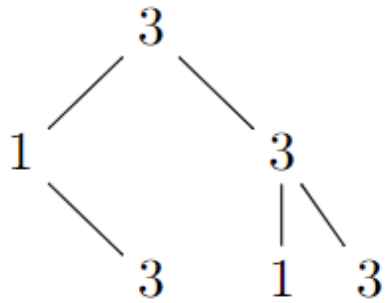
Example: Pictorial Way to See $P(i), I(i)$



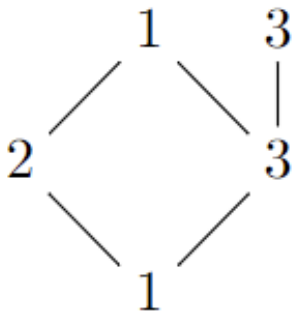
$P(1)$



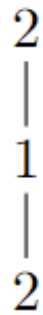
$P(2)$



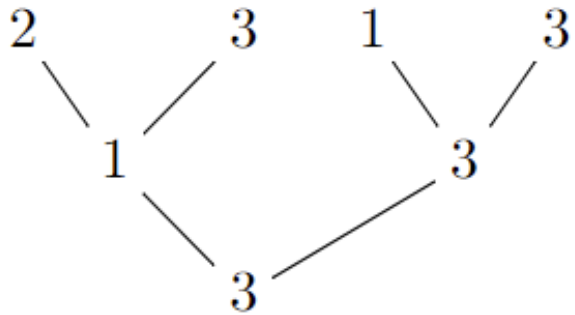
$P(3)$



$I(1)$



$I(2)$

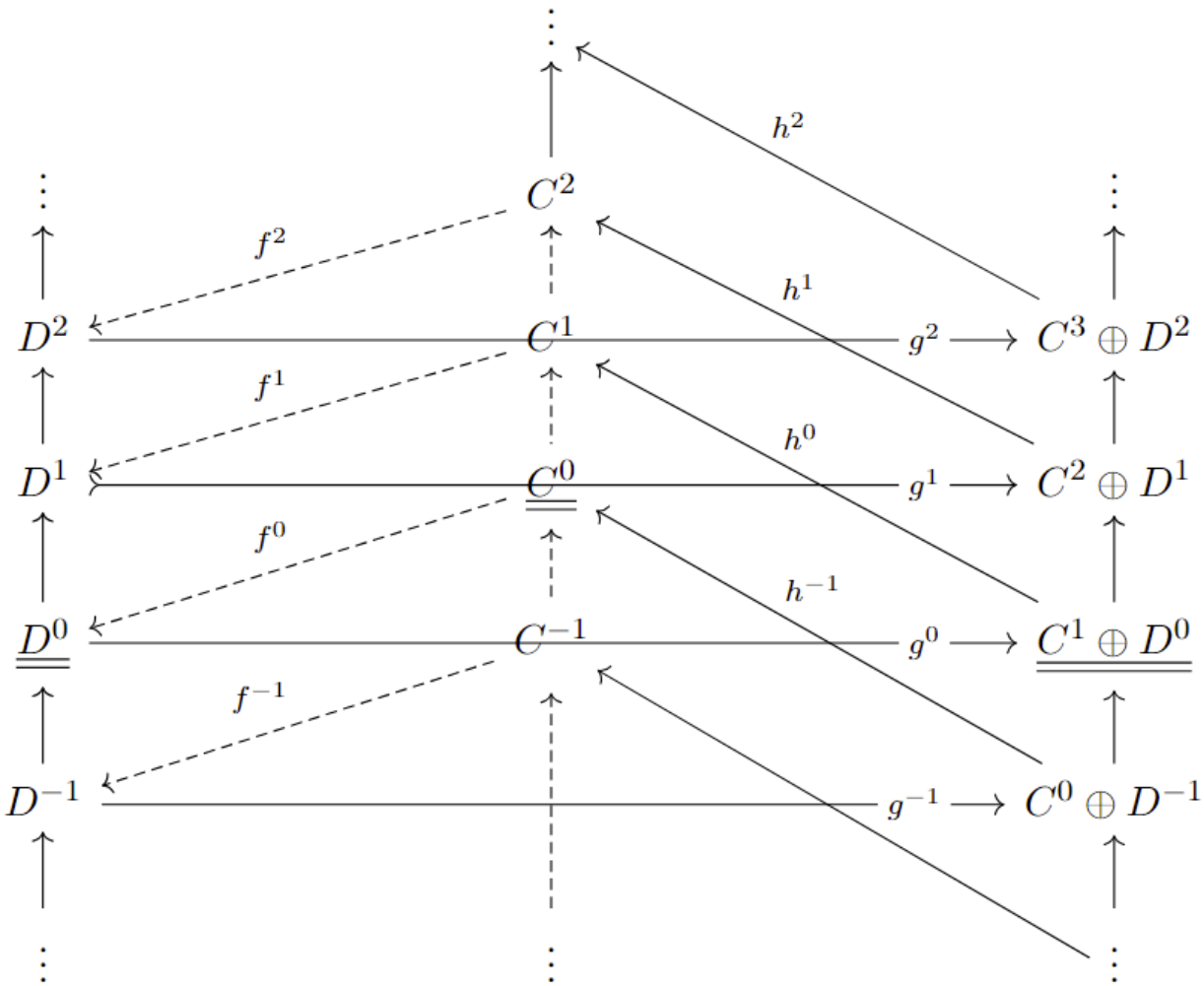


$I(3)$

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Motivation



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Motivation (2)

- Slogan: Work with *triangles* instead of short exact sequence.
- We define the notion of *triangulated category*: it is a category with the extra information of $[1]$ and a class of **distinguished triangles**.
- In the category of chain complexes (and the derived category), the distinguished triangle are of the above form.

Definition

A **triangle** in $\mathscr{C}$ is a ordered triple of (cochain) maps (f,g,h), with $X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} X[1]$.

Definition

Given two triangles $X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} X[1]$, $X' \xrightarrow{f'} Y' \xrightarrow{g'} Z' \xrightarrow{h'} X'[1]$, a **morphism of triangles** in $\mathscr{C}$ between them is a triplet (α,β,γ) such that the diagram

$$\begin{array}{ccccccc} X & \xrightarrow{f} & Y & \xrightarrow{g} & Z & \xrightarrow{h} & X[1] \\ \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \beta} & & \downarrow{\scriptstyle \gamma} & & \downarrow{\scriptstyle \alpha[1]} \\ X' & \xrightarrow{f'} & Y' & \xrightarrow{g'} & Z' & \xrightarrow{h'} & X'[1] \end{array}$$

is commutative in $\mathscr{C}$. We say two triangles are **isomorphic** if all α,β,γ are isomorphisms in $\mathscr{C}$.

The Correct Definition?

There are two things we want:

1. (Rotation Axiom.) If $X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} X[1]$ is distinguished, then so is $Z[-1] \xrightarrow{h[-1]} X \xrightarrow{f} Y \xrightarrow{g} Z$.
2. $0 \to X \xrightarrow{f} Y \xrightarrow{g} Z \to 0$ short exact, then exists $X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} X[1]$ distinguished.

The Correct Definition?

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It turns out that we can only have both theorems in $\mathscr{D}(R)$.

Definition (Informal)

The **derived category** of a ring R, denoted by $\mathscr{D}(R)$, is a category with objects as chain complexes over R-module, and morphisms as formal symbols fs^{-1}, where s is a quasi-isomorphism.

Triangulated Subcategories

Definition

A full subcategory L is a **triangulated subcategory** of a category $\mathscr{C}$ if the following conditions holds:

- If X is in L, then $X[n]$ also a complex of L for all $n \in \mathbb{Z}$.
- If $X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} X[1]$ is a distinguished triangle in $\mathscr{C}$, and X, Y are objects in L, then Z is also in L.

Corollary

Given L, a triangulated subcategory in $\mathscr{D}(R)$ and $0 \to X \xrightarrow{f} Y \xrightarrow{g} Z \to 0$ short exact sequence. If any two of X, Y, Z is in L, then so is the remaining one.

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What is... Injective Generation?

Given a collection of (R-module) chain complexes $\mathscr{S}$.

Definition

We denote $\langle \mathscr{S} \rangle$ to be the smallest full triangulated subcategory of $\mathscr{D}(R)$ which contains $\mathscr{S}$ (as objects) and is closed under coproduct.

What is... Injective Generation?

Given a collection of (R-module) chain complexes $\mathscr{S}$.

Definition

We denote $\langle \mathscr{S} \rangle$ to be the smallest full triangulated subcategory of $\mathscr{D}(R)$ which contains $\mathscr{S}$ (as objects) and is closed under coproduct.

Then consider all injective (resp. projective) R-modules $\mathrm{Inj}\text{-}R$ (resp. $\mathrm{Proj}\text{-}R$)

Definition

If $\langle \mathrm{Inj}\text{-}R \rangle = \mathscr{D}(R)$ we say **injective generates** for R.

Showing Injective Generation: Technique

Theorem

Consider the following bounded above exact complex of R-modules:

$$\cdots \longrightarrow C_n \cdots \longrightarrow C_{n-1} \longrightarrow \cdots \longrightarrow C_0 \longrightarrow X \longrightarrow 0 \longrightarrow 0 \longrightarrow \cdots \tag{1}$$

and suppose $C_n, C_{n-1} \cdots, C_0$ are all modules in $\langle \text{Inj-}R \rangle$, then X is also in $\langle \text{Inj-}R \rangle$.

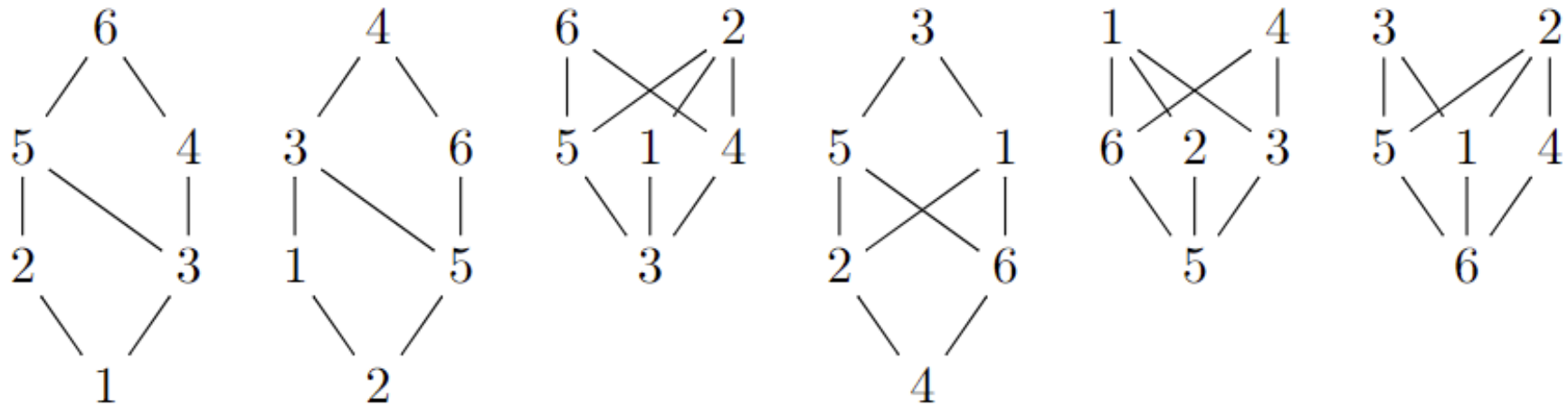
And in the case of $A = kQ/I$,

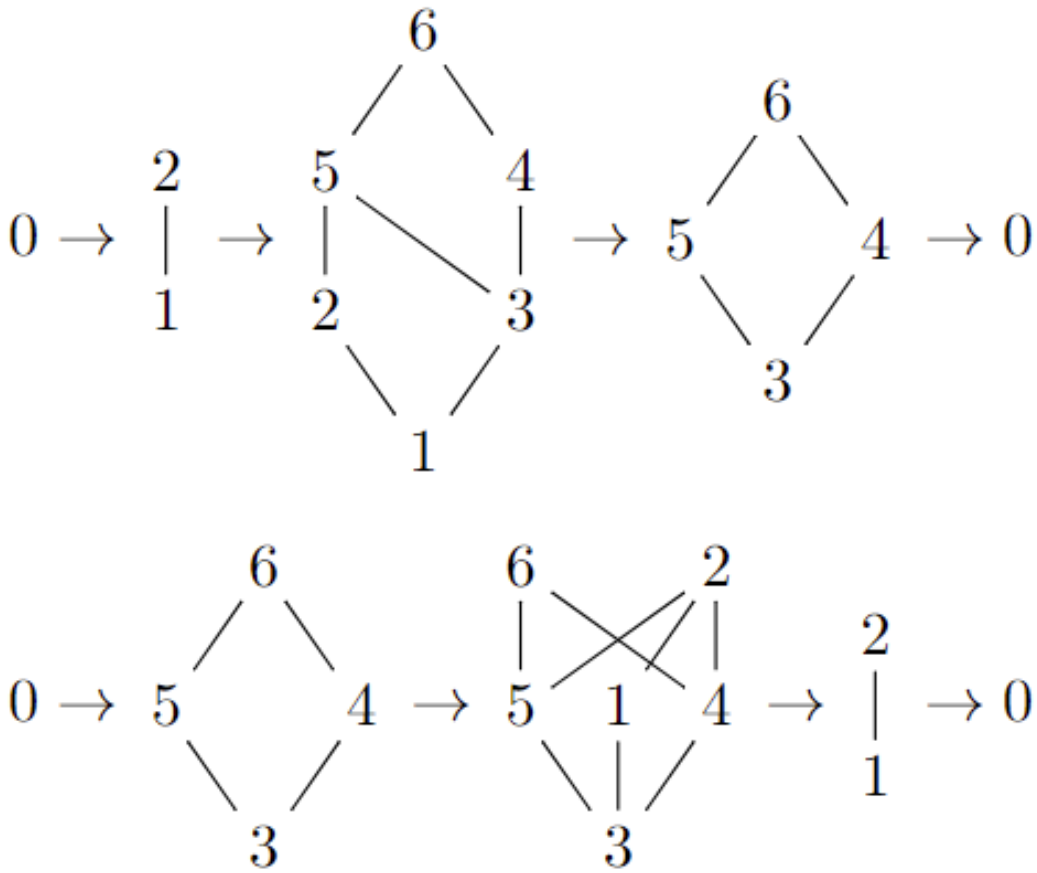
Theorem

If Q has n vertices and $S(1), S(2), \cdots S(n) \in \langle \text{Inj-}R \rangle$, then injective generates for A.

Showing Injective Generation: Example

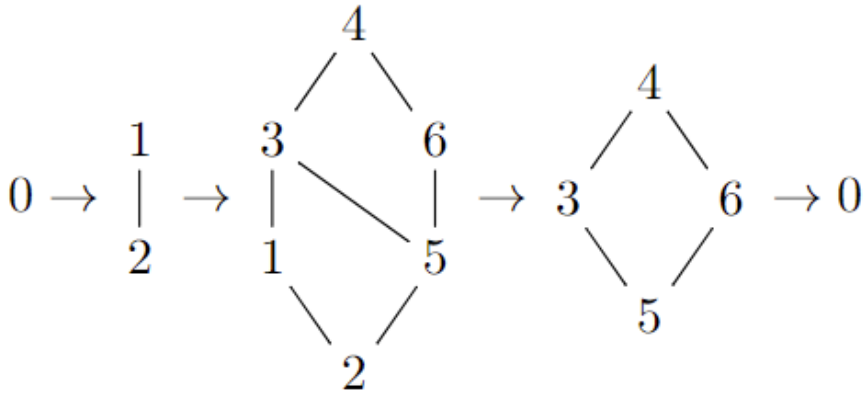
The following modules are $I(1), \cdots I(6)$:

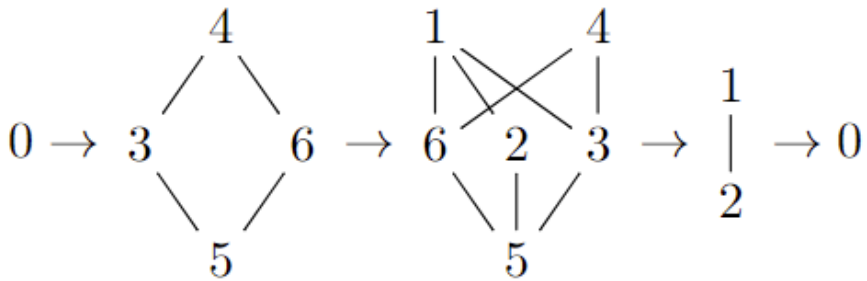



0 → 2 → 5 4 → 6 → 6 5 4 → 0
1 2 3 1 3
0 → 6 5 4 → 6 2 5 1 4 → 2 → 0
3 3 1

$$\cdots \to I(3) \to I(1) \to I(3) \to I(1) \to \begin{array}{ccc} & 6 & \\ \diagup & & \diagdown \\ 5 & & 4 \\ \diagdown & & \diagup \\ & 3 & \end{array} \to 0$$

$$\cdots \to I(1) \to I(3) \to I(1) \to I(3) \to \begin{array}{c} 2 \\ | \\ 1 \end{array} \to 0,$$





$$0 \to 1 \cong S(1) \to \begin{matrix} 2 \\ | \\ 1 \end{matrix} \to 2 \cong S(2) \to 0, \text{ and } 0 \to 2 \to \begin{matrix} 1 \\ | \\ 2 \end{matrix} \to 1 \to 0$$

Thank You!

Any questions?